INCH-POUND
MIL-DTL-12109/21A
20 August 1998
SUPERSEDING
MIL-S-12109/21
31 December 1976

DETAIL SPECIFICATION SHEET

SOLENOID, ELECTRICAL; CLASS 2 (MACHINE GUN)
7.62 MM, 24 VOLT DC, 8 AMP MAX

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-12109 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

APPLICABLE DOCUMENTS:

Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-C-13486	- Cable, Special Purpose, Electrical: Low-Tension, Heavy-Duty, Single-Conductor and Multiconductor, Shielded and Unshielded, General Specification For.
MIL-R-3065	- Rubber, Fabricated Products.

STANDARDS

DEPARTMENT OF DEFENSE

NASM20995	- Wire, Safety or Lock.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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MS24677	- Screw, Cap, Socket Head, Hexagon, Drilled Alloy Steel (UNC-3A).
NASM33540	- Safety Wiring and Cotter Pinning, General Practices For.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

7792103	- Gasket.
7792104	- Cable Assembly.
75020922	- Connector Assembly.

(Application for copies should be addressed to the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issue of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI J STD 004,005,006	- Flux, Soldering, Liquid, Paste Flux, Solder Paste, and Solder-Paste Flux; General Specification For.

(Application for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

REQUIREMENTS:

Requirements for submission of first article shall be as specified in the contract.

Stroke shall be 0.165 to 0.247 inches (in.) (0.42 to 0.63 centimeters (cm)).

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Weight shall not exceed 2.25 pounds (lb) (1 kilogram (kg)).

Solenoid shall functionally operate against the static forces specified in figure 1, at 20 and 30 volts direct current (Vdc).

The current shall not exceed 8 amperes maximum.

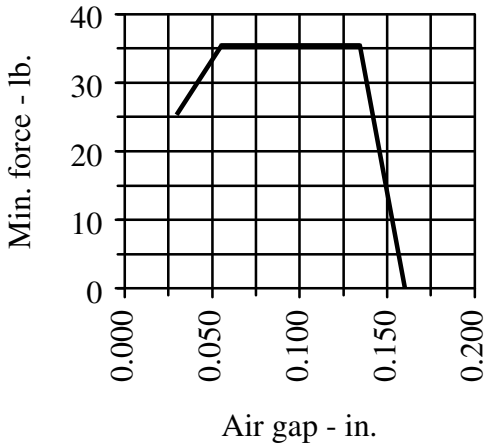

FIGURE 1. Static force - air gap curve.

Solenoid shall meet the static force air gap curve shown in figure 1.

Solenoid shall be capable of firing 10 000 rounds.

Solenoid shall withstand a dielectric withstanding voltage potential of 1000 volts root mean square. A leakage current of 100 micro-amperes max is permitted.

VERIFICATION:

The first article shall be submitted for inspection in accordance with contract requirements. The first article shall be tested in accordance with the tests specified herein and the testing provisions of MIL-DTL-12109.

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a. The stroke test shall be performed during functional (operating) test.
b. Weight shall be confirmed by any approved method.
c. Testing for dielectric withstanding voltage shall as follows:

Test voltage of 1000 volts root mean square at 60 Hertz (Hz) in frequency and shall approximate, as closely as possible, a true sine wave in form. The voltage shall be applied between coil terminal and ground metal parts for at least 60 seconds. Test voltage shall be raised from zero to 60 Hz in frequency at a rate of approximately 500 volts per second.

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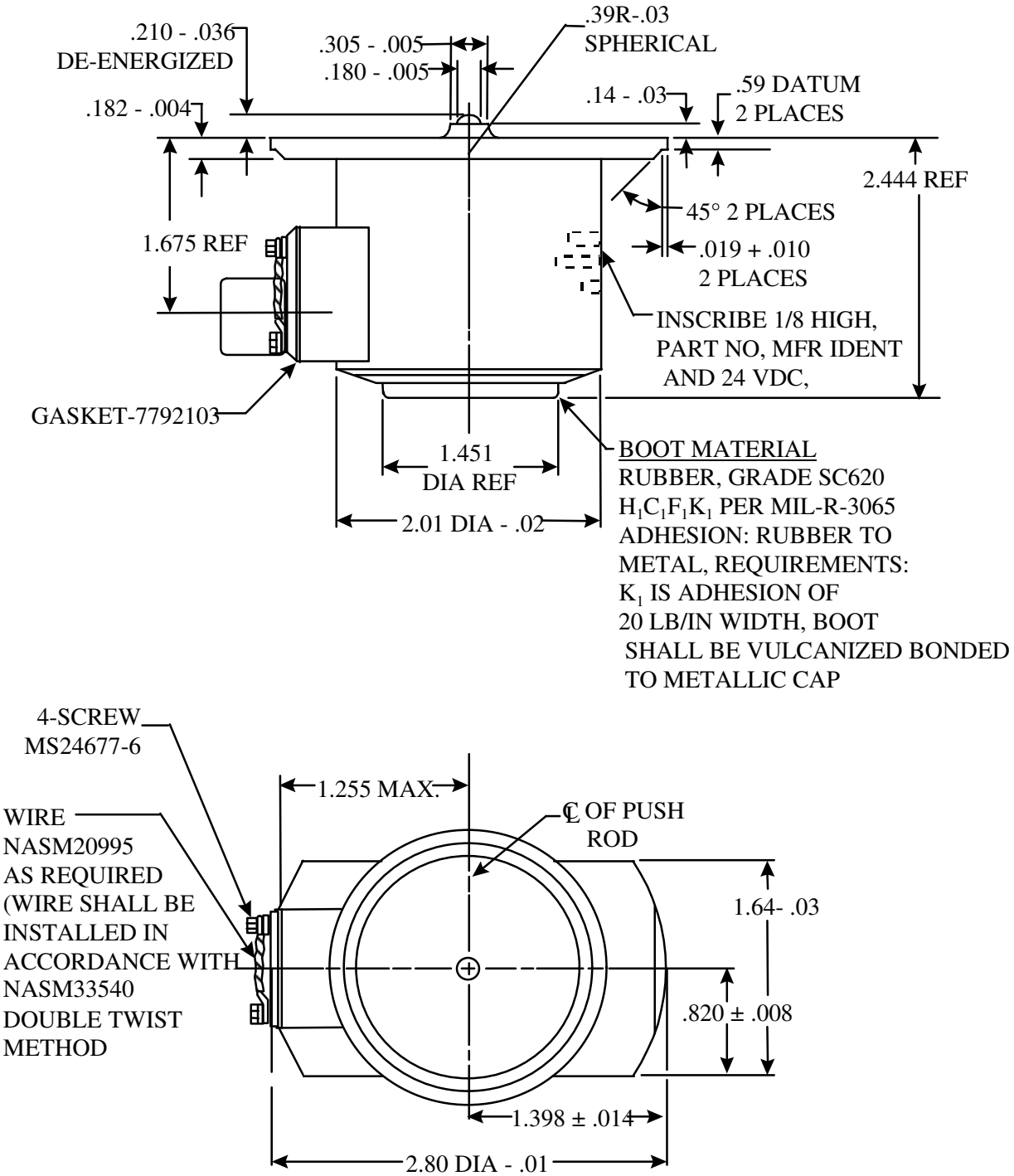

FIGURE 2. Solenoid characteristics.

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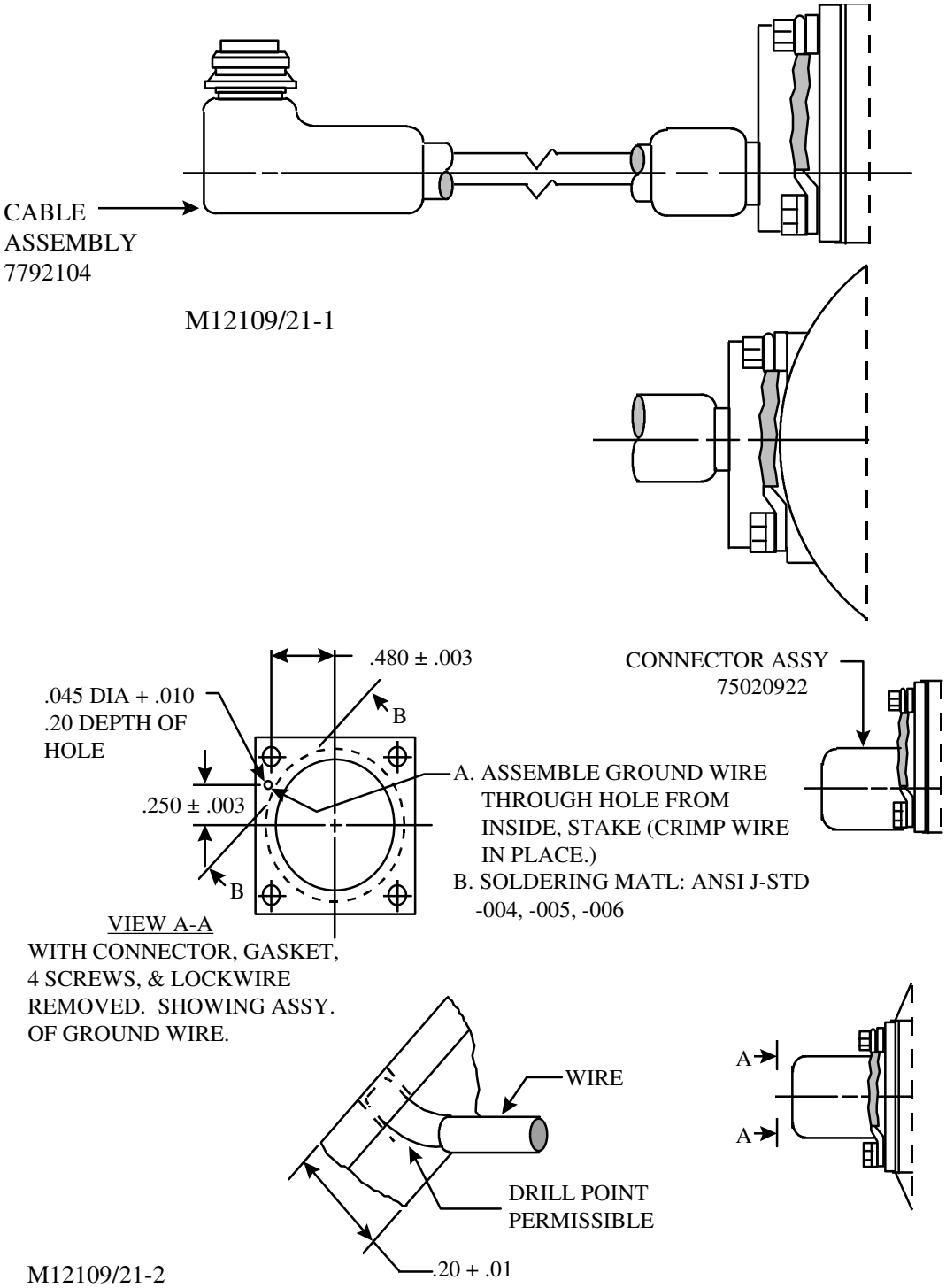

FIGRE 2. Solenoid characteristics - Continued.

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NOTES:
1. This drawing is for interface and envelope requirements and is not intended to limit design.
2. Dimensions are in in. Unless otherwise specified, tolerances are ± .010 on decimals, ± 2° on angular dimensions.
3. REF = Reference.

FIGURE 2. Solenoid characteristics - Continued.

NOTES:

PART OR IDENTIFYING NUMBERS (PIN):

M12109/21-1 (Former part number 7791685 with weapon designation M60C).
M12109/21-2 (Former part number 75F20924 with weapon designation M60E2).

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - AT
Navy - OS
Air Force - 85

Preparing Activity:
Army - AT

(Project 5945-1025-04)

Review Activities:
Army - AR, EA
Navy - MC
Air Force - 99
DLA - CC

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